ADAM SHAW

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PROFESSIONAL SUMMARY

Third-year undergraduate at USC pursuing a B.S. in Applied Mathematics and M.S. in Statistics, with research experience in topological deep learning. Currently investigating how topological simplification affects learning and bidirectional capability in predictive coding networks. Seeking PhD opportunities in Machine Learning or Statistics to continue research in neural network learning dynamics.

SKILLS

- Python
- PyTorch
- Scikit-learn
- Pandas
- Seaborn
- C++

EDUCATION

University of Southern California | Los Angeles, CA — Expected to Graduate December 2027

B.S., Applied Mathematics; Computer Science Minor
M.S., Statistics
Cumulative GPA: 3.7/4.0

Relevant Coursework (All A's):
Mathematics of Machine Learning, Mathematical Statistics, Data Science with Python, Graduate Analysis of Algorithms

EXPERIENCE

Student Researcher — August 2025 - Present
University of Southern California | Los Angeles, CA

- Conducting research with Professor Alvin Jin in the field of topological deep learning
- Analyzing changes in topological invariants (Betti numbers) across network layers and their effect on inversion ability in predictive coding networks, a type of invertible neural network
- Presenting findings at the Conference on Topological Data Analysis at the University of Missouri in November and preparing manuscript for submission

Associate Software Developer Intern — May 2025 - August 2025
Google | New York, NY

- Joined the Workspace AI Platform team, which integrates Gemini into products such as Google Docs, Slides, and Gmail
- Designed and built an access control system for custom Gems (personalized Gemini instances), streamlining creation and secure sharing across Google Workspace teams
- Implemented role-based permissions in C++ using Google's internal server framework, managing authorization and routing for millions of potential users

Student Researcher — December 2024 - May 2025
University of Southern California | Los Angeles, CA

- Worked with Professor James Alcala on the identification and classification of bias in text containing hyperbolic and sarcastic language
- Developed a multidimensional bias classification system to quantify political/ideological bias in hyperbolic and sarcastic text